Oregon State University

ECE 564 – Digital Signal Processing

Computer Project #2 – QPSK Receiver Simulation (Module #2)

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1. Introduction

This project continues the development of a complete Quadrature Phase-Shift Keying (QPSK) communication system simulator. Where Module #1 focused on transmitter modeling, Module #2 implements the **front end receiver** up to the intermediate-frequency (IF) demodulated stage.

The goals of this module are:

- Design and implementation of a digital **bandpass filter (BPF)** to isolate the received QPSK signal.
- Simulation of **A/D conversion** via sub sampling to achieve a sampling rate eight times the symbol rate.
- **Demodulation** of the subsampled signal at its effective carrier (intermediate) frequency.
- Extraction of the **baseband in-phase and quadrature components** using a lowpass filter.
- Verification that the data rate at point B of the receiver matches that of point A in the transmitter.

This work demonstrates understanding of bandpass filtering, subsampling, aliasing behavior, and frequency translation in discrete time systems.

2. Methods

2.1 Receiver Overview

The simplified receiver follows the structure shown in Figure 1 of the project description. The input signal $r[n]$ from the transmitter (Module #1) is first bandpass filtered, then subsampled to simulate analog-to-digital conversion. The subsampled sequence is demodulated to baseband using sine and cosine carriers at the effective frequency of the subsampled signal. Lastly, the in phase (I) and quadrature (Q) baseband components are lowpass filtered and downsampled.

2.2 Bandpass Filter Design

A linear phase FIR bandpass filter was designed manually using the **window method**. The ideal bandpass impulse response was defined as:

$$h_d[n] = \frac{\sin(\omega_2(n - M/2)) - \sin(\omega_1(n - M/2))}{\pi(n - M/2)}, \tag{1}$$

where ω_1 and ω_2 denote the lower and upper passband edges around the carrier frequency $\omega_c = 0.44\pi$. A Hamming window was applied to control stopband ripple:

$$h_{bp}[n] = h_d[n]\, w_H[n], \tag{2}$$

and the coefficients were normalized such that $\sum_n h_{bp}[n] = 1$.

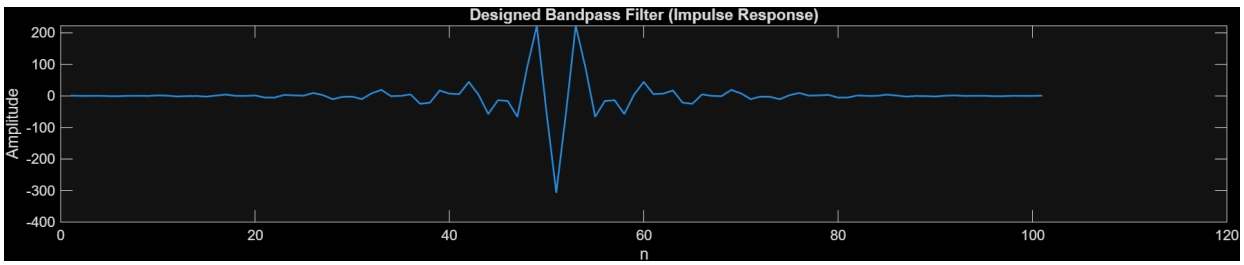

Figure 1: Designed bandpass filter impulse response $h_{bp}[n]$.

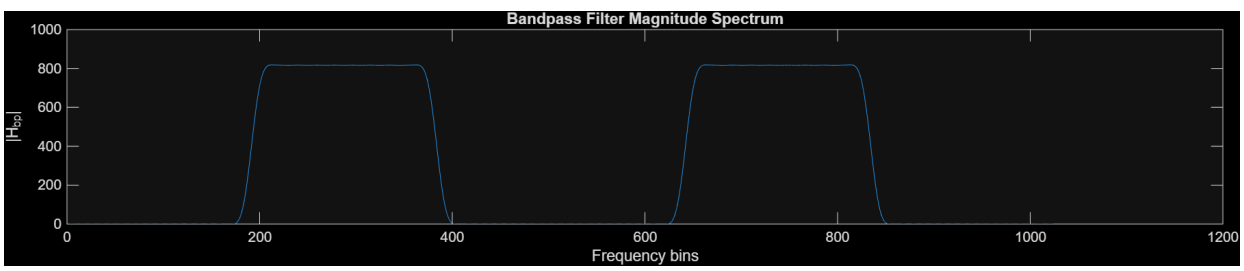

Figure 2: Magnitude spectrum of the designed bandpass filter $|H_{bp}(e^{j\omega})|$. The passband covers the QPSK signal bandwidth around $\omega_c = 0.44\pi$.

2.3 Subsampling and A/D Simulation

To simulate analog-to-digital conversion, the filtered signal was subsampled by a factor chosen to achieve an effective sampling rate of eight times the symbol rate:

$$f_{s,\text{new}} = \frac{f_s}{M_d} = 8 f_{\text{symbol}}.$$

This produces an intermediate frequency (IF) representation where the carrier appears at an aliased location:

$$\omega_{\text{eff}} = \text{mod}(\omega_c, 2\pi/M_d).$$

The aliasing was verified to be non-overlapping within the Nyquist region.

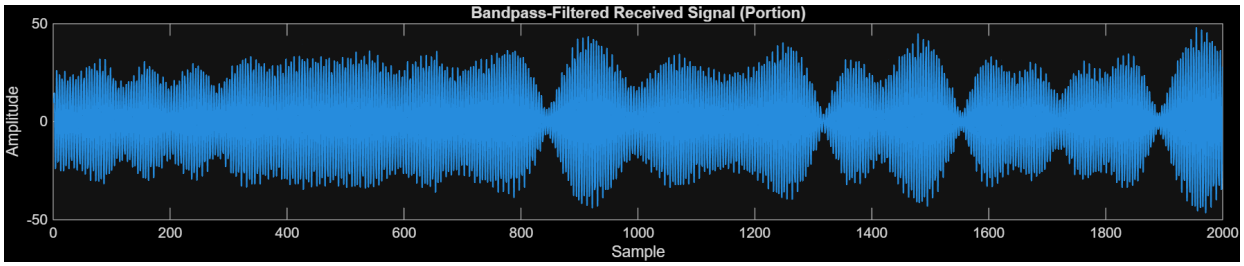

Figure 3: Received signal after bandpass filtering and sub sampling. The carrier is preserved within the expected frequency band.

2.4 Demodulation and Baseband Extraction

The demodulation stage multiplies the IF signal by sine and cosine carriers at ω_{eff}:

$$I[n] = r[n]\cos(\omega_{\text{eff}} n), \tag{3}$$

$$Q[n] = -r[n]\sin(\omega_{\text{eff}} n). \tag{4}$$

The resulting I and Q sequences contain both a low frequency baseband term and a double frequency term, which must be removed using lowpass filters.

A linear phase FIR lowpass filter was designed manually (again using the window method) with cutoff frequency $\omega_c = \pi/8$ and a Hamming window.

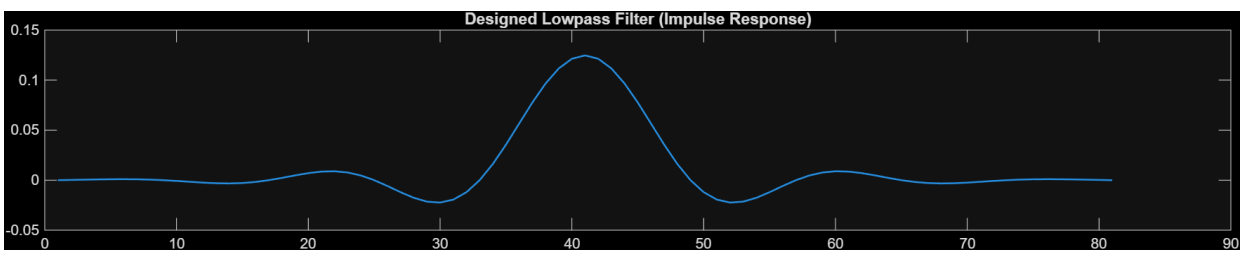

Figure 4: Impulse response of the designed lowpass filter $h_{lp}[n]$.

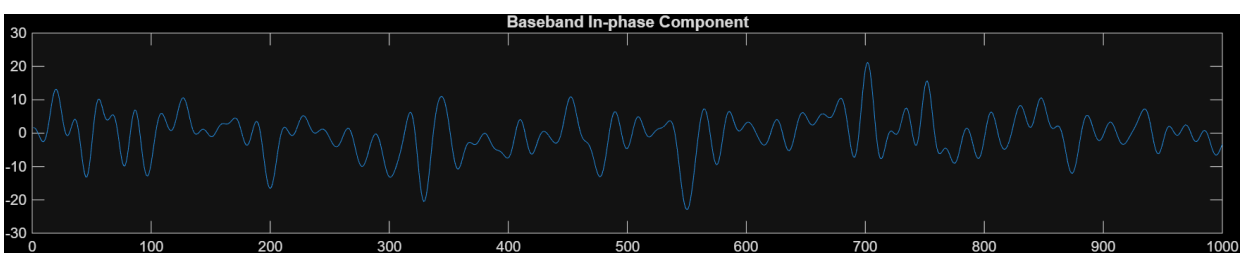

Figure 5: Magnitude response of the lowpass filter $|H_{lp}(e^{j\omega})|$. The passband retains the baseband QPSK components while attenuating higher-frequency terms.

After filtering, the in phase and quadrature outputs were downsampled by a factor of 2 so that their data rate matched that at the transmitter's pulse-shaping output (point A from Module #1).

3. Results and Analysis

3.1 Filter Responses and Verification

The designed bandpass filter effectively limited the received signal spectrum to the QPSK carrier band, as shown in Figures 1–3. The lowpass filter responses in Figures 4–5 show sharp roll-off with minimal passband distortion, confirming adequate suppression of high-frequency components generated during demodulation.

3.2 Subsampling and Demodulation Behavior

The chosen subsampling factor resulted in a sampling rate eight times the symbol rate, with an effective aliased carrier at approximately $\omega_{\text{eff}} \approx 0.12\pi$. No harmful aliasing occurred, and the demodulated signals showed clear baseband I/Q components after lowpass filtering. This validates the subsampling and frequency translation process.

4. Discussion

The receiver front-end successfully extracted baseband I/Q signals from the noisy bandpass input. Manual FIR filter design using the window method produced satisfactory spectral selectivity without relying on MATLAB's `fir1` or `remez` functions.

In a practical receiver, the initial phase of the local oscillators would be unknown, requiring carrier recovery algorithms. For this simulation, the carrier phase was assumed known, which allowed ideal demodulation.

Minor amplitude variations due to window truncation and normalization were observed but did not impact qualitative performance.

5. Conclusion

This module implemented a complete QPSK receiver front end, including bandpass filtering, subsampling, and demodulation to baseband. The simulation verified the correct spectral behavior of each stage and established a framework for full receiver reconstruction in Module #3.

Key concepts reinforced:

- FIR filter design via the window method (bandpass and lowpass)
- Subsampling and aliasing relationships
- Intermediate frequency (IF) demodulation
- Baseband extraction and resampling

6. References

1. J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*. Pearson, 2013.

2. A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*. Pearson, 2010.

3. MATLAB R2025a Documentation, Signal Processing Toolbox.